
Preprocessing & Feature Engineering Log

Module: IT3091 – Machine Learning
Project: Ames Housing Price Prediction

1. Outlier Removal

- **Action:** Filtered the dataset to remove rows where $\text{GrLivArea} > 4000$ AND $\text{SalePrice} < 300000$.
- **Result:** Two extremely large, low-priced outlier records were successfully dropped from the training set to prevent model distortion.

2. Feature Engineering

Created three new composite features based on domain knowledge of real estate valuation:

1. TotalSF (Total Square Footage)

- **Formula:** $\text{TotalBsmtSF} + \text{1stFlrSF} + \text{2ndFlrSF}$
- **Rationale:** Buyers often value the total usable space of a house rather than treating basement and above-ground space entirely separately. Consolidating this reduces multicollinearity.

2. PropertyAge

- **Formula:** $\text{YrSold} - \text{YearBuilt}$
- **Rationale:** The absolute year built (e.g., 1995) is less intuitively meaningful to a model than the actual age of the property at the time it was sold (e.g., 12 years old).

3. TotalBaths

- **Formula:** $\text{FullBath} + (0.5 * \text{HalfBath}) + \text{BsmtFullBath} + (0.5 * \text{BsmtHalfBath})$

- **Rationale:** Aggregates all bathroom data into a single numerical metric representing the total bathroom count of the home, which is a standard real estate listing feature.

3. Data Imputation Strategy

Implemented via `sklearn.compose.ColumnTransformer`:

- **Numerical Features (SimpleImputer):**
 - **Strategy:** median
 - **Reason:** Fills missing values with the median of the column (fitted only on training data) to avoid the influence of extreme outlier values.
- **Categorical Features (SimpleImputer):**
 - **Strategy:** constant, `fill_value='None'`
 - **Reason:** Missing values in features like PoolQC or Fence imply the absence of the feature rather than missing data. 'None' is treated as a valid category.

4. Feature Scaling and Encoding

Implemented within the same `ColumnTransformer` pipeline:

- **Numerical Features (StandardScaler):**
 - Scales all numerical features to have a mean of 0 and a standard deviation of 1. While tree-based models don't strictly require scaling, it is necessary for the LinearRegression benchmark and ensures numerical stability.
- **Categorical Features (OneHotEncoder):**
 - **Parameter:** `handle_unknown='ignore'`
 - Converts categorical text into binary matrix columns. If a category appears in the validation set that wasn't in the training set, it is ignored (all zeros) rather than causing a pipeline crash.

5. Pipeline Integration

- **Action:** The Imputer, Scaler, and Encoder were bundled into a Pipeline structure.
- **Benefit:** This strictly enforces that all `fit()` operations (e.g., finding the median, calculating the standard deviation) are executed *only* on the training split, completely eliminating the risk of data leakage from the validation set.